24/9/2026

Practical - 8

a) Nmap Live Host Discovery ARP, ICMP, and TCP/UDP Ping Scans.

Aim:

To discover live hosts on a network using Nmap host-~~dive~~ discovery scans - ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

Introduction

Nmap ("Network Mapper") is a free, open-source network-scanning tool created by Gordon Lyon (Fyodor) and first released in 1997. It is the industry standard tool for mapping networks, discovering live host, and identifying the services running on them.

Before Nmap can usefully scan ports on a target it first needs to know whether the target is even online. Port-scanning an offline address wastes time and creates unnecessary network noise - so host discovery is always the first practical step of a network assessment.

Nmap can determine whether a host is alive using four different protocols, one at each relevant layer of the TCP/IP stack:

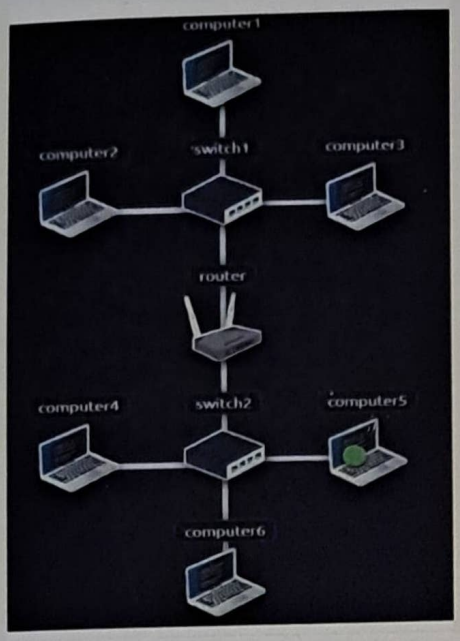
ARP (Link Layer) - for hosts on the same local subnet.

ICMP (Network Layer) - echo, timestamp, and address mask requests.

TCP (Transport Layer) - SYN and ACK ping probes

UDP (Transport Layer) - relies on an ICMP "port unreachable" reply.

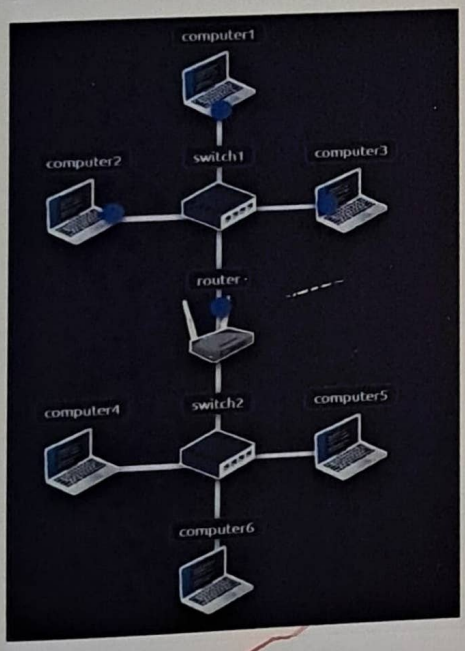
otocol Nmap Uses for



TCP & UDP ping scans
(-PS / -PA / -PU)

ICMP ping scans (-PE / -PP / -PM)

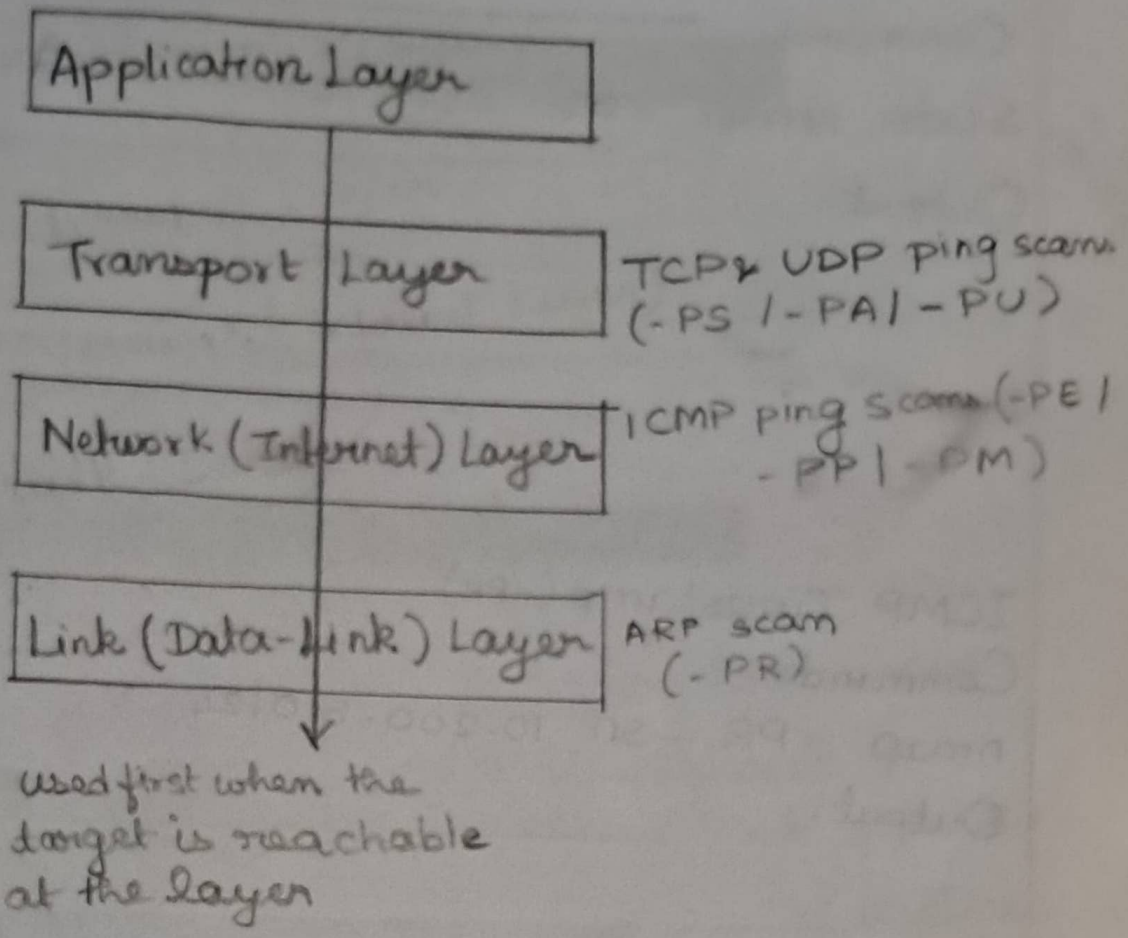
ARP scan
(-PR)



00.6.0/24

.6.0/24
6-09-28 05:58 UTC
in 206.13 seconds

TCP/IP Layers and the Protocol Nmap Uses for Host Discovery



ARP Host Discovery:

Command:

```
sudo nmap -PR -sn 10.200.6.0/24
```

Output:

```
root@ip-10-48-167-237:~# sudo nmap -PR -sn 10.200.6.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-28 05:58 UTC
Nmap done: 256 IP addresses (0 hosts up) scanned in 206.13 seconds
```

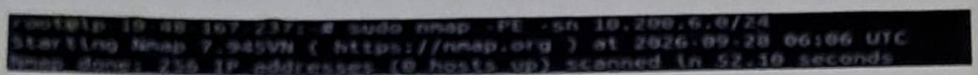
ICMP Host Discovery

ICMP Echo (-PE)

Command:

sudo nmap -PE -sn 10.200.6.0/24

Output:



ICMP Timestamp (-PP)

Command:

nmap -PR -sn 10.200.6.0/24

Output:

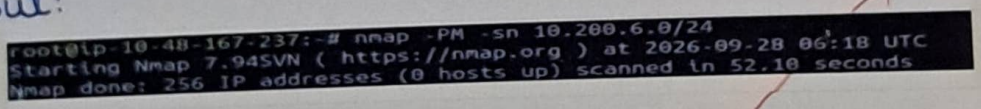


ICMP Address Mask (-PM)

Command:

nmap -PM -sn 10.200.6.0/24

Output:



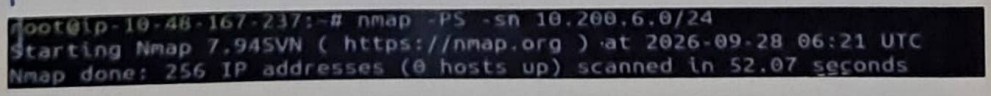
TCP and UDP Ping Discovery

TCP SYN Ping (-PS)

Command:

nmap -PS -sn 10.200.6.0/24

Output:



TCP ACK Ping (-PA)
Command:
sudo nmap -PA -sn 10.200.6.0/24
Output:

```
root@ip-10-48-167-237:~# sudo nmap -PA -sn 10.200.6.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-28 06:24 UTC
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.05 seconds
```

UDP Ping (-PU)
Command
sudo nmap -PU -sn 10.200.6.0/24
Output:

```
root@ip-10-48-167-237:~# sudo nmap -PU -sn 10.200.6.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-28 06:26 UTC
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.34 seconds
```

Nmap's Default Host-Discovery Behaviour
Local network: Nmap uses ARP by default.
Remote network: Nmap uses ICMP and TCP probes.
-sn → Host discovery only
-Pn → Skip host discovery and treat hosts as online
Masscan - A Faster Alternative
Command:
masscan 10.200.6.0/24 -p443
Output:

```
root@ip-10-48-167-237:~# masscan 10.200.6.0/24 -p443
Command 'masscan' not found, but can be installed with:
apt install masscan
```

Command:
masscan 10.200.6.0/24 -p80,443
Output:

```
root@ip-10-48-167-237:~# masscan 10.200.6.0/24 -p80,443
Command 'masscan' not found, but can be installed with:
apt install masscan
```

Command:
masscan 10.200.6.0/24 -p22 -25
Output:

```
root@ip-10-48-167-237:~# masscan 10.200.6.0/24 -p22-25
Command 'masscan' not found, but can be installed with:
apt install masscan
```

Reverse DNS Lookup
rDNS: Resolves an IP address to ~~be~~ a hostname
-R → Forces DNS lookup
-n → Disables DNS lookup
--dns-servers → Specifies a DNS server

Command:
nmap -R -sn 10.200.6.0/24
Output:

```
root@ip-10-48-167-237:~# nmap -R -sn 10.200.6.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-28 06:44 UTC
Nmap done: 256 IP addresses (0 hosts up) scanned in 210.28 seconds
```

For disalfling DNS lookups:
Command:
nmap -n -sn 10.200.6.0/24
Output:

```
root@ip-10-48-167-237:~# nmap -n -sn 10.200.6.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-28 06:48 UTC
Nmap done: 256 IP addresses (0 hosts up) scanned in 206.11 seconds
```

Result:
Thus, live hosts were successfully discovered using Nmap, ARP, ICMP, and TCP/UDP ping scans, and the different host-discovery techniques and their corresponding network layers were understood.

27/9/26

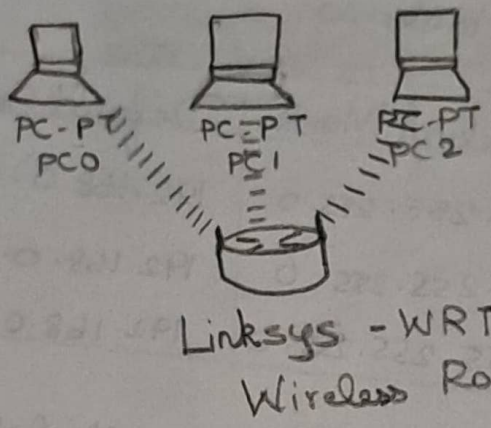
Practical 8

b) Configuration of wireless LAN using Cisco Packet Tracer

Aim:

To do configuration of wireless LAN using Cisco Packet Tracer.

Design a topology with three PCs connected from Linksys Wireless routers.



Perform the following configuration:

* Configure Static IP on PC and Wireless Router
* Set SSID to MotherNetwork
* Set IP address of router to 192.168.0.1, PC0 to 192.168.0.2, PC1 to 192.168.0.3 and PC2 to 192.168.0.4
* Secure your network by configuring WAP key on Router
* Connect PC by using WAP key

To complete these tasks follow these step by step instructions.

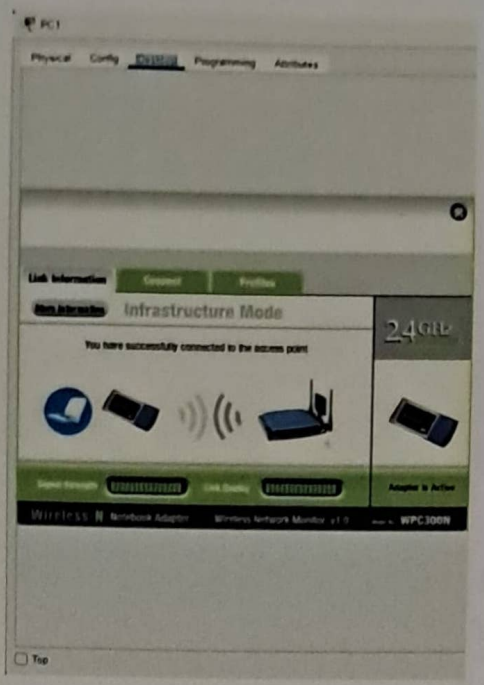
Step 1: Click on wireless router

* Select Administration tab from top Menu, set username and password to admin and click on Save Setting.
* Next click on wireless tab and set default SSID to MotherNetwork.
* Now select wireless security and change Security Mode to WEP.

* Set key1 to 0123456789
* Again go in the end of page and Click on Save Setting
* Now we have completed all given task on Wireless router. Now configure the static IP on all three PC's
* Double click on the PC select Desktop tab click on IP configuration select Static IP and set IP as given below.

PC	IP	Subnet Mask	Default Gateway
PC0	192.168.0.2	255.255.255.0	192.168.0.1
PC1	192.168.0.3	255.255.255.0	192.168.0.1
PC2	192.168.0.4	255.255.255.0	192.168.0.1

* Now its time to connect PCs from Wireless router. To do so click PC select Desktop click on PC Wireless.
* Click on connect tab and click on Refresh button
* It will ask for WAP key insert 0123456789 and click connect.
* It will connect you with wireless router.
* The system is connected. And PC1 cord is active.
* Repeat same process on PC1 and PC2.

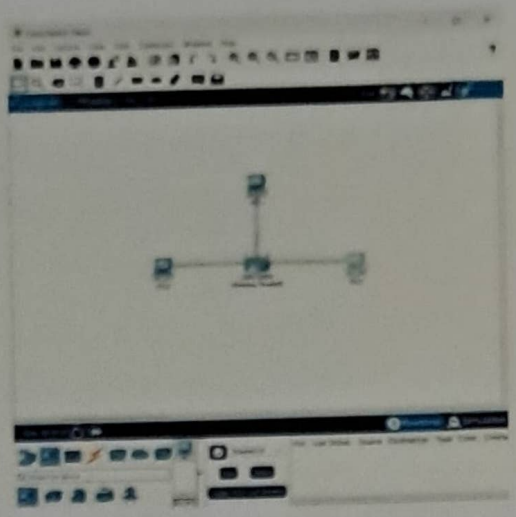



```
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=29ms TTL=255
Reply from 192.168.0.1: bytes=32 time=10ms TTL=255
Reply from 192.168.0.1: bytes=32 time=11ms TTL=255
Reply from 192.168.0.1: bytes=32 time=13ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 29ms, Average = 15ms
```



Result:
Thus configuration of Wireless LAN using Cisco Packet Tracer is successfully implemented.